Shivam Raj

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SKILLS

Languages:	C, C++, Java, Python, JavaScript, TypeScript
Backend:	Spring Boot, Node.js, Express, FastAPI, REST APIs, JWT Authentication
Frontend:	HTML, CSS, React.js
Database:	MongoDB, PostgreSQL, pgvector
Machine Learning:	Scikit-learn, Computer Vision, Feature Engineering, ML pipelines, RAG, Local LLMs (Ollama)
Tools:	Git, Docker, Linux, AWS EC2, Swagger/OpenAPI
Frameworks:	Node, Spring
Soft Skills:	Articulation, Organizer, Critical Thinking, Leadership

EXPERIENCE

AI Lead | Aispire Pvt Ltd | Vyrix — Jun' 2026 - Present

- Built the AI for **Vyrix**, a product of Aispire Pvt Ltd that gives design students a **tailored research assistant**.
- Developed the product's **local AI** to help students **validate assumptions**, **spot gaps**, and decide the next step in their projects.

PROJECTS

CampusGPT | GitHub — Apr' 2026

- Built a **private RAG study assistant** using **Spring Boot** and React that answers questions from uploaded PDFs with a **local LLM** on Ollama.
- Implemented **hybrid retrieval** combining **pgvector semantic search** and full-text keyword search using **Reciprocal Rank Fusion** and MMR re-ranking.
- Developed **embedding analytics** in pure Java with **K-Means clustering**, **PCA**, and **LOF anomaly detection**.
- Added 6 study modes, **SSE streaming**, **JWT authentication**, rate limiting, and source citations with PDF page numbers.
- **Tech Stack:** React, TypeScript, Spring Boot, PostgreSQL, pgvector, Ollama, Docker

WriteFlow | GitHub — Apr'2026 - Present

- Built a **full-stack blog platform backend** using **Spring Boot** with a **decoupled REST API architecture** supporting **posts, comments, and user management**.
- Designed **secure backend services** with **JWT-based authentication**, **role-based access control (RBAC)**, **rate limiting**, and **input validation** following **OWASP best practices**.
- Implemented features including **pagination**, **profile management**, and **content workflows** using **Spring Data JPA** with **optimized database handling** (**PostgreSQL/H2**).
- Deployed the application using **Docker-based containerization** with **Nginx reverse proxy**, enabling **seamless frontend-backend integration** and **scalable system design**.
- **Tech Stack:** React, Spring Boot, PostgreSQL, Java, JWT, Docker, Nginx

RepMate | GitHub — Jun'2025 - Present

- Built a **real-time fitness tracking system** using **computer vision** (MediaPipe) to detect body keypoints and use **machine learning** to classify exercise form with **probabilistic scoring**.
- Designed a **feature engineering pipeline** from **pose keypoints** (joint angles, velocity, temporal smoothing) for **sequence-based** rep detection and form evaluation.
- Integrated a **FastAPI-based ML inference module** with **WebSocket communication** into a **Spring Boot backend** with **JWT-based authentication** for low-latency real-time feedback.
- Implemented analytics features including rep tracking, performance metrics, and gamification, achieving **~96% accuracy** in controlled test scenarios.
- **Tech Stack:** React, Spring Boot, PostgreSQL, Python, FastAPI, MediaPipe, OpenCV, Scikit Learn

Gym Management | GitHub | Project Link — Feb' 2026

- Built a **full-stack gym management platform** using **Spring Boot** and React with a decoupled **REST API architecture** for handling member lifecycle, subscriptions, and administrative workflows.
- Designed secure backend services with **JWT-based authentication**, **role-based access control (RBAC)**, rate limiting, and input validation to ensure system security.
- Implemented core modules including membership management, trainer allocation, attendance tracking, and financial records with **PostgreSQL** and optimized query handling.
- Deployed the application using **containerized** backend services and cloud hosting, with API documentation via **Swagger** and real-time dashboard analytics supported by **caching mechanisms**.
- **Tech Stack:** React, Spring Boot, PostgreSQL, Java, JWT

FinTrackAI | GitHub | Project Link Jul' 2025

- Built a full-stack financial analytics platform using the **MERN stack** with secure **JWT-based authentication** and **Google OAuth** for user management.
- Developed automated expense tracking and data processing pipelines supporting **CSV/PDF ingestion**, validation, and structured storage using **MongoDB**.
- Integrated **AI-driven financial insights** using external models and designed a **classification pipeline** using text features (**TF-IDF**) with scope for ML-based expense categorization.
- Implemented **interactive dashboards** for real-time visualization of spending patterns, savings trends, and categorized financial insights.
- **Tech Stack:** MongoDB, Express.js, React.js, Node.js, Scikit Learn, REST APIs

TRAINING

FLAMES '25 MERN with Gen AI | W3grads | Certificate Jul'2025

- Resolved development bottlenecks in **AI workflows** and database performance by optimizing pipelines and debugging inconsistencies, improving **system reliability** and deployment stability.
- Improved team efficiency by establishing a structured **Git workflow** and enforcing **daily SCRUMs**, reducing merge conflicts and ensuring consistent collaboration across distributed contributors.
- Enhanced application scalability by **addressing deployment and integration challenges** in **full-stack systems**, enabling smoother releases and faster iteration cycles.
- Built and refined a **full-stack MERN application (FinTrackAI)** during training, contributing to **frontend UI/UX development**, **API integration**, and **end-to-end feature implementation**.
- Integrated **Gemini API-based AI features** and validated backend services using **Postman**, ensuring **robust API performance**, **accurate data flow**, and **reliable system behavior**.
- Engineered **AI-enabled features using Gemini API** and optimized **API testing workflows with Postman**, improving **debugging efficiency** and ensuring **high reliability of backend services**.

CERTIFICATES

- Master Generative AI & Generative AI tools (Infosys) | Certificate Aug' 2025
- Build Generative AI Apps and Solutions with No-Code Tools | Certificate Aug' 2025
- Java Programming (iamneo) | Certificate May' 2025
- Data Structures and Algorithms with C++ (iamneo) | Certificate Dec' 2024
- The Bits and Bytes of Computer Networking (Google) | Certificate Sep' 2024
- Computer Communications (University of Colorado) | Certificate Sep' 2024
- TCP/IP and Advanced Topics (University of Colorado) | Certificate Sep' 2024

ACHIEVEMENT

- Solved 500+ DSA Problems on LeetCode, including 130+ Hard Nov' 2026
- MEDHA 2024 Hackathon: Achieved 5th rank among 60 teams. Aug' 2024

EDUCATION

Lovely Professional University Phagwara, Punjab

- Bachelor of Technology - Computer Science and Engineering – CGPA: 8.34 Aug' 2023 – Present

Touch Wood School Dehradun, Uttarakhand

- Intermediate – Percentage: 80 Apr' 2022 – Mar' 2023
- Matriculation – Percentage: 90 Apr' 2020 – Mar' 2021